# Skills Summary

## Dividing Complex Numbers

*Use this reference while completing your worksheet or when studying.*

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## Skill 1 — Dividing by a Pure Imaginary Number

**Conjugate:** the conjugate of $ci$ is $-ci$, and $(ci)(-ci) = c^2$, a real number.

**Method:** multiply the top and the bottom by that conjugate, replace every $i^2$ with $-1$, then divide each part of the numerator by the real denominator.

**Example:** Simplify $\dfrac{12+8i}{4i}$

**Step 1.** The denominator is a single imaginary term, so the fastest conjugate to use is $-4i$ — it clears the $i$ from the bottom in one step.

**Step 2.** Top: $(12+8i)(-4i) = -48i + 32$. Bottom: $(4i)(-4i) = 16$. Dividing both parts by 16 gives the real part 2 and the imaginary part $-3$.

$\Rightarrow$ $\mathbf{2 - 3i}$

**Try it:** Simplify $\dfrac{15-10i}{5i}$

check: $\mathbf{-2 - 3i}$

## Skill 2 — Dividing by a Complex Binomial

**Conjugate:** the conjugate of $c+di$ is $c-di$, and their product is always real:

$$(c+di)(c-di) = c^2 + d^2$$

That real value is the new denominator, which is why the conjugate — and not the divisor itself — is the thing to multiply by.

**Example:** Simplify $\dfrac{34}{5-3i}$

**Step 1.** An $i$ in the denominator means the quotient is not in standard form yet, so multiply top and bottom by the conjugate $5+3i$.

**Step 2.** Bottom: $25 + 9 = 34$. Top: $34(5+3i)$. The 34 cancels, leaving the real part 5 and the imaginary part 3.

$\Rightarrow$ $\mathbf{5 + 3i}$

**Try it:** Simplify $\dfrac{20+10i}{1+2i}$

check: $\mathbf{8 - 6i}$

## Skill 3 — Quotients in an Applied Setting

**Where this shows up:** in an alternating-current circuit, voltage, current and impedance are complex, and *current* = *voltage* ÷ *impedance*. Divide exactly as above; the real part of the answer is the in-phase amount and the imaginary part records the phase shift. Keep the unit that belongs to the quotient — volts ÷ ohms gives amps.

**Example:** A voltage of $15+5i$ volts drives an impedance of $1+2i$ ohms. Find the current in amps.
**Step 1.** Current is voltage divided by impedance, so this is a quotient of two complex numbers — multiply top and bottom by the conjugate $1-2i$.
**Step 2.** Bottom: $1+4=5$. Top: $(15+5i)(1-2i)=25-25i$. Dividing both parts by 5 gives the real part 5 and the imaginary part $-5$.
$\Rightarrow$ $\mathbf{5-5}\boldsymbol{i}$ amps

**Try it:** A voltage of $22+4i$ volts drives an impedance of $1+i$ ohms. Find the current in amps.

check: $\mathbf{13-9}\boldsymbol{i}$ amps

**(!) Watch out:** multiply by the *conjugate*, not by the denominator itself — multiplying $c+di$ by $c+di$ leaves an $i$ downstairs. And $(c+di)(c-di)=c^2+d^2$, a *sum*: the minus sign turns into a plus when $i^2$ is replaced by $-1$.